# MADHAV SEMWAL

## Software Engineer | Backend & API Development

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Backend-focused Software Engineer with hands-on experience in Node.js, REST APIs, WebSockets, OAuth 2.0, and Database. Skilled in designing scalable systems, secure authentication services, and developer-focused tools with an emphasis on performance and maintainability.

## WORK EXPERIENCE

**SDE Intern,** *Sequoia Automation and Controls Pvt.Ltd* **09/2026 – Present | Remote**

- Developed **RESTful APIs** using **Node.js**, architecting database schemas in **PostgreSQL** for reliable data storage.
- Streamlined the **CI/CD deployment pipeline** using **GitHub Actions and Docker**, ensuring backend code passed **automated unit tests** before containerization.
- Decreased **PostgreSQL database load** by implementing a **Redis caching strategy** for frequently accessed data, significantly improving **API response times** for core endpoints.

**Web Developer,** *Freelance* **2024 – Present | Remote**

- Built and deployed **3 customized e-commerce catalog systems** for freelance clients, integrating **Node.js REST APIs** with **React.js and Tailwind CSS** frontends.
- Optimized **backend data delivery** by removing unused fields from **JSON payloads** and indexing **MongoDB queries**, reducing **data transfer** and improving **frontend load times**.

## PROJECT

**AuthSphere** , *authentication platform* **11/2025 – Present**

- Developed a centralized **OAuth 2.0 authentication provider** using **Node.js, TypeScript, and MongoDB**, featuring **PKCE** and **JWT refresh token rotation**.
- Developed and tested a reusable **TypeScript SDK** using **Jest**, enabling plug-and-play **authentication** across web applications with **90%+ test coverage**.
- Designed **RESTful APIs** for **authorization and token exchange**, integrating **WebSockets** to broadcast **real-time session state** changes and enable instant cross-device logouts

**AICOMM** , *AI-Powered Git Commit Assistant* **02/2025 – Present**

- Built a **Node.js CLI** that analyzes native **Git diffs** to dynamically prompt the **Gemini API** for **Conventional Commit messages**, designing **fallback routing strategies** that maintain **100% CLI reliability**.
- Integrated native **Git status and diff parsing** to automatically extract and summarize **staged and unstaged code changes**.
- Designed a lightweight developer interface featuring customizable configurations, a dry-run execution mode, and automated commit generation.

**Go File Organizer** , *Concurrent CLI Utility* **February 2026**

- Built a **concurrent Go CLI utility** to automatically categorize files using a **bounded worker pool**, **Goroutines**, and **buffered channels** for asynchronous processing.
- Implemented **thread-safe file operations** using sync/atomic and native **OS packages**, with **timestamp-based collision handling** and a **dry-run mode** to prevent data loss.

## SKILLS

**Languages**
JavaScript, TypeScript, Go, C++

**Backend & Databases**

- Node.js, Express.js, PostgreSQL, MongoDB, Redis

**Frontend Technologies**
React.js, Next.js, HTML5, CSS3, Tailwind CSS (or standard UI libraries)

**DevOps & Tools**
Docker, GitHub Actions (CI/CD), Jest, Go Test, Git/GitHub, REST APIs, WebSockets

## EDUCATION

**B.E. in Computer (Ongoing),** *Savitribai Phule Pune University* **Aug 2023 – 2027 | Pune, India**
Data Structures & Algorithms (implemented in C++), Operating Systems, Database Management Systems, Computer Networks, Software Engineering.